

Linux Cheat Sheet for Beginners

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Introduction to Linux:

- Linux is an open-source operating system kernel used by various distributions (distros) like Ubuntu, Debian, CentOS, etc. It's widely used in servers, embedded systems, and personal computers.
- The terminal is a text-based interface where users can interact with the system by typing commands.

Useful Commands to interact with Linux system

Here’s the updated **Ubuntu Linux Cheat Sheet for Beginners**, now with examples for each command:

File Management

Command	Description	Example
ls	List files and directories in the current directory.	ls -l (detailed view)
cd <directory>	Change the current directory.	cd /home/user/Documents
pwd	Print the current working directory.	pwd
cp <source> <destination>	Copy files or directories.	cp file.txt /tmp/
mv <source> <destination>	Move or rename files and directories.	mv file.txt newfile.txt
rm <file>	Remove a file.	rm file.txt
rm -r <directory>	Remove a directory and its contents.	rm -r old_folder
cat <file>	Display file contents.	cat notes.txt

<code>less <file></code>	View file contents one page at a time.	<code>less bigfile.log</code>
<code>head <file></code>	View the first 10 lines of a file.	<code>head data.csv</code>
<code>tail <file></code>	View the last 10 lines of a file.	<code>tail data.csv</code>
<code>touch <file></code>	Create an empty file or update timestamp.	<code>touch newfile.txt</code>
<code>find <path> -name <filename></code>	Search for files by name.	<code>find /home -name "*.txt"</code>

Folder Management

Command	Description	Example
<code>mkdir <directory></code>	Create a new directory.	<code>mkdir new_folder</code>
<code>rmdir <directory></code>	Remove an empty directory.	<code>rmdir empty_folder</code>
<code>tree</code>	View directories in a tree structure.	<code>tree /var/log</code>

Disk Management

Command	Description	Example
<code>df -h</code>	Show disk space usage in human-readable format.	<code>df -h</code>
<code>du -sh <path></code>	Show size of a directory or file.	<code>du -sh /home/user</code>
<code>mount <device> <mountpoint></code>	Mount a disk or partition.	<code>sudo mount /dev/sdb1 /mnt</code>

<code>umount <mountpoint></code>	Unmount a disk or partition.	<code>sudo umount /mnt</code>
<code>lsblk</code>	List information about block devices.	<code>lsblk</code>
<code>fdisk -l</code>	Show partition table of disks.	<code>sudo fdisk -l</code>

Package Management

Command	Description	Example
<code>sudo apt update</code>	Update package list.	<code>sudo apt update</code>
<code>sudo apt upgrade</code>	Upgrade installed packages.	<code>sudo apt upgrade</code>
<code>sudo apt install <package></code>	Install a package.	<code>sudo apt install vim</code>
<code>sudo apt remove <package></code>	Remove a package.	<code>sudo apt remove vim</code>
<code>dpkg -l</code>	List installed packages.	<code>`dpkg -l</code>
<code>dpkg -i <package.deb></code>	Install a <code>.deb</code> package.	<code>sudo dpkg -i mypackage.deb</code>

User Management

Command	Description	Example
<code>whoami</code>	Display the current user.	<code>whoami</code>
<code>sudo adduser <username></code>	Add a new user.	<code>sudo adduser john</code>

<code>sudo passwd <username></code>	Change a user's password.	<code>sudo passwd john</code>
<code>sudo deluser <username></code>	Delete a user.	<code>sudo deluser john</code>
<code>groups <username></code>	Display groups a user belongs to.	<code>groups john</code>

Process Management

Command	Description	Example
<code>ps aux</code>	Show all running processes.	<code>`ps aux</code>
<code>top</code>	Monitor system processes in real time.	<code>top</code>
<code>htop</code>	Enhanced process viewer.	<code>htop</code>
<code>kill <PID></code>	Kill a process by PID.	<code>kill 1234</code>
<code>killall <name></code>	Kill processes by name.	<code>killall firefox</code>
<code>bg</code> and <code>fg</code>	Resume jobs in the background or foreground.	<code>bg</code> (background), <code>fg</code> (foreground)

Networking

Command	Description	Example
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<code>ip a</code>	Display network interfaces.	<code>ip a</code>
<code>ping <host></code>	Test network connectivity to a host.	<code>ping google.com</code>
<code>curl <url></code>	Fetch content from a URL.	<code>curl http://example.com</code>
<code>wget <url></code>	Download files from a URL.	<code>wget http://example.com/file.zip</code>
<code>netstat -tuln</code>	Show open ports and connections.	<code>netstat -tuln</code>
<code>ss -tuln</code>	Display open ports and connections.	<code>ss -tuln</code>

General Troubleshooting

Command	Description	Example
<code>dmesg</code>	Display kernel messages.	<code>`dmesg</code>
<code>journalctl -xe</code>	View detailed system logs.	<code>journalctl -xe</code>
<code>systemctl status <service></code>	Check the status of a service.	<code>systemctl status ssh</code>
<code>sudo reboot</code>	Reboot the system.	<code>sudo reboot</code>
<code>sudo shutdown now</code>	Shut down the system immediately.	<code>sudo shutdown now</code>
<code>uptime</code>	Display system uptime.	<code>uptime</code>
<code>free -h</code>	Show memory usage.	<code>free -h</code>

Kubernetes Basics (Linux Context)

Command	Description	Example
<code>kubectl get nodes</code>	List all nodes in the cluster.	<code>kubectl get nodes</code>
<code>kubectl get pods</code>	List all pods in the cluster.	<code>kubectl get pods -n kube-system</code>
<code>kubectl describe pod <pod-name></code>	Show detailed information about a pod.	<code>kubectl describe pod nginx</code>
<code>kubectl logs <pod-name></code>	Fetch logs for a pod.	<code>kubectl logs nginx</code>
<code>kubectl exec -it <pod-name> -- bash</code>	Execute a command inside a pod.	<code>kubectl exec -it nginx -- bash</code>
<code>kubectl apply -f <file.yaml></code>	Apply a Kubernetes manifest file.	<code>kubectl apply -f deployment.yaml</code>

This comprehensive cheat sheet provides clear examples to help beginners quickly understand and use the commands.